

BETA-HP-200 Hydraulic Press - Equipment Manual

Field	Value
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Tenant	Beta Manufacturing (tenant-beta)
Site	Harbor Works, Line 1 (imperial documentation)
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Equipment	BETA-HP-200
Also covers	none; the HP-200S straight-side variant is out of scope, see Documentation gaps
Document type	Equipment manual
Revision	Rev B
Effective date	2025-05-15
Status	current
Supersedes	BETA-MAN-HP200-A
Applies to	Assets listed under equipment
Related documents	BETA-SAF-HP200-ECP-1, BETA-PM-HP200-B, BETA-DIA-P100-1
Document owner	Maintenance Engineering

Purpose

This manual describes the design, rated limits and safe operation of the BETA-HP-200 H-frame hydraulic press, 160 tons, installed on Line 1 at Harbor Works.

It is the controlled reference used when building a work order for this press.

All values in this manual are in imperial units: psi, inches, lbf.ft and gpm. Do not convert a value into metric and act on the converted figure. A converted figure is not a documented figure and the work order must record the imperial value and the revision it came from.

Reason for this revision

Rev B was issued after the Q1 2025 Line 1 incident review and the 2025 reliability review. The incident review found one clamp circuit failure that released a tool under load and one light curtain event caused by a badly fitted replacement emitter.

The changes in Rev B are conservative: pressures are lower, the die entry speed is lower, the hold-to-run time is longer and the inspection intervals are shorter. Nothing in Rev B relaxes a limit.

Equipment overview

The BETA-HP-200 is a single H-frame press with a side mounted hydraulic power unit, a 25 mm die cushion on the ram and a light curtain across the front of the die area. Tooling is clamped by two hydraulic tool clamps on the lower bolster.

The die cushion stores energy: it is a gas or oil filled cylinder that is compressed when the ram descends through the last inch of stroke. The cushion

pressure must be released and the cushion cylinder blocked before anyone enters the die area.

Guard interlocks are on the front sliding guard. The press cannot cycle with any guard open, and muting of the light curtain is prohibited on this machine.

Technical data

Parameter	Value
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Rated press force	160 tons at 32 in daylight
Relief valve set point, main circuit	1750 psi
Maximum permissible working pressure	1650 psi
Hard-wired high pressure trip	1850 psi
Minimum hydraulic clamp pressure	1800 psi
Rated capacity daylight	32 in
Stroke	24 in
Die cushion stroke	1 in
Die closing height	6 in minimum, 46 in maximum
Bed size	42 in x 34 in, T-slots on 4 in centres
Ram speed, approach	4 in/s
Ram speed, working	2 in/s
Ram speed at die entry	3 in/s maximum for daylight above 40 in, otherwise 5 in/s
Motor	25 hp, 460 V, three phase
Pump	duplex, 30 gpm, two pumps P-100A and P-100B
Hydraulic oil	ISO VG 46 anti-wear hydraulic oil, Beta specification BM-H46
Reservoir	180 gal, sight glass and float switches
Return line filtration	10 micron
Tie-rod nut torque	2200 lbf.ft, re-torque every 1500 running hours
Tie-rod re-torque method	three passes in diagonal sequence
Die height sensor calibration	every 2000 running hours
Pressure gauge calibration	every 2000 running hours
Hydraulic oil analysis	every 500 running hours, and after any hose failure
Ram parallelism	0.008 in maximum per 12 in of die face
Ram parallelism check	every 1000 running hours
Sound pressure at the operator station	88 dB(A) during the working stroke

Running hours are read from the hour meter at the operator station. The hour meter counts only while the pump motor is energised.

Operating conditions

Ambient air temperature: 40 degF to 104 degF. The press must not be started with oil temperature below 50 degF; warm the oil with the pumps in bypass. Hydraulic oil temperature: 68 degF to 131 degF continuous. Indoor, non-corrosive service only. Do not use the press in the wash-down bay or outdoors. Permitted operations: stamping, coining, rivet setting and plastic compression. Forging, welding and any operation that puts sparks near the seals is prohibited. Maximum tool weight is 500 lb. Heavier tools are handled with the BETA-OC-5T crane using a spreader beam. Sound levels above 85 dB(A) require hearing protection, which is posted at the press and included in the line hearing protection plan.

Safeguards and control system

Light curtain across the front of the die area, safety rated, category 3, PL d. Muting is prohibited on this press. There is no muting function in the control software and enabling one is an unauthorised modification.

Safety-rated stop time, light curtain beam to ram stop: 200 ms maximum.

Hold-to-run time: 1000 ms minimum. Releasing the cycle button before 1000 ms must not produce any ram motion.

Two independent ram position limit switches plus an anti-two-block contact.

Tool clamps must be at or above the minimum clamp pressure in Technical data before the ram can descend. The interlock is defeated when the pressure switch does not prove; a press that will not hold clamp pressure must not be used.

The light curtain must never be defeated, taped over, blocked or bypassed with an opaque object. Any diagnostic that requires defeating the safety circuit requires a documented risk assessment, a written permit and a second authorised person present.

Every replacement emitter, column or receiver must be function tested before the press is returned to production. An emitter that has been replaced and not function tested is an undocumented safety device.

Operating limits and permitted range

1. Working pressure must not exceed 1650 psi at the manifold. The relief valve is set at 1750 psi and is not adjustable in service. A reading above 1650 psi with an empty die is a stop condition: stop the press, do not adjust the relief valve, and raise a work order.
2. Pressing force is limited to 160 tons at 32 in daylight. The displayed force falls off as the die closes because the transducer is calibrated for full daylight.
3. Point loading outside the bolster T-slots is prohibited.
4. The die cushion must be at its rated pressure before the ram is lowered.
5. Ram parallelism must be within 0.008 in per 12 in of die face before a precision operation.
6. Any event in which the hard-wired high pressure trip operated is a structural inspection trigger.

Variant coverage

Parameter	BETA-HP-200	BETA-HP-200S straight-side	
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Rated force	160 tons	160 tons	
Frame	H-frame	Straight-side	
Daylight	32 in	32 in	
Hydraulic clamp pressure, minimum	1800 psi	1800 psi	
Sound pressure at the operator station	88 dB(A)	85 dB(A)	
Guard arrangement	sliding front guard, light curtain	fixed guard, light curtain	

BETA-HP-200S is serviced under work instruction WI-2204. That instruction is not part of this document collection. If the asset plate carries the suffix S, stop and raise a documentation request before any work on the clamp circuit, the guard or the cushion.

BETA-HP-250 is a different machine with a 250 ton rating and different limits. BETA-HP-100 is a bench press in the maintenance shop. Neither is covered here.

Maintenance and inspection requirements

The preventive maintenance programme is in BM-PM-HP200. The limits from this manual that the programme must satisfy:

- Hydraulic oil analysis every 500 running hours, and after any hose failure.
- Tie-rod re-torque every 1500 running hours, three diagonal passes.
- Die height sensor calibration every 2000 running hours.
- Pressure gauge calibration every 2000 running hours.
- Ram parallelism check every 1000 running hours.
- Structural inspection of the frame welds every 3000 running hours, and after any overload event.
- Reduced die entry speed for tools with daylight above 40 in.

Documentation gaps

The following values are not contained in this manual. They are held in the as-built records, which are not part of this document collection:

- Die cushion gas or oil charge pressure for this serial number: on the commissioning sheet CM-HP200-01.
- Clamp cylinder and cushion cylinder seal part numbers.
- Manifold and tube fitting torque values: on the fitting vendor data sheets.
- The work instruction WI-2204 that covers the HP-200S straight-side variant.
- The light curtain manufacturer service procedure, which covers the response time calculation for muting requests; muting is prohibited here in any case.

Escalation conditions

Escalate immediately, before continuing the diagnosis, when any of the following occurs:

- Hydraulic pressure above 1650 psi with no tooling load in the die.
- An audible knock from the pump that does not stop when the relief opens.
- Visible hydraulic oil in the die area, on the tie rods or inside the frame.
- Ram side movement above 0.08 in, or a parallelism reading outside 0.008 in per 12 in.
- Hydraulic clamp pressure below the minimum in Technical data.
- Any crash of the light curtain during a cycle.
- Any overload event in which the high pressure trip operated.
- Any repeated trip of the safety circuit on the same cycle.
- A required value is missing from the controlled document set.

Escalation route: Level 2 Maintenance Engineering, then the Line 1 Supervisor, then EHS for anything involving stored energy or structural damage.

Appendix: Revision B change impact assessment

Revision B changed the relief set point and the oil analysis interval from Revision A. This appendix records the impact so that a technician holding an older work order can see what must be re-checked.

Change	Revision A	Revision B	Impact
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Main relief set point	2000 psi	1750 psi	Reduce the setting and re-seal
Oil analysis interval	500 running hours	300 running hours	Update the PM schedule
Tie-rod re-torque interval	4000 running hours	2000 running hours	Update the PM schedule
Accumulator pre-charge	72 psi	72 psi	No change

The relief valve change is the significant one. A press still set to 2000 psi after the revision takes effect will exceed the maximum permissible working pressure of the circuit and is a stop condition. The setting is verified at every annual service and after any relief valve work.

Work orders raised before the Revision B effective date remain valid until worked; a work order that specifies a superseded value must be reissued against Revision B before the work starts.

Appendix: Operator daily and weekly checks

Daily and weekly checks are the first line of defence against a failure found later at a planned service. They are performed by the operator and countersigned by the shift supervisor.

Daily checks before first start:

Confirm the hydraulic oil level is between the low and high marks.
Confirm the oil temperature is above 68 degF before full load.
Confirm the guard and light curtain are in place and undamaged.
Confirm the emergency stop is not latched and resets correctly.
Confirm there is no oil on the floor or the bolster.
Confirm the pressure gauge reads zero with the pump stopped.

Weekly checks:

Confirm the gauge matches the controller within 70 psi.
Confirm the tie-rod nuts show no movement against the witness marks.
Confirm the gib bolts are tight and the ram moves without hesitation.
Confirm the oil is clear and free of milky emulsion.
Confirm the accumulator safety block is open and tagged.
Confirm the pump and motor mounts are not loose.

Any check that fails is reported before the next shift. A failed safety check stops the press until corrected and re-verified.

Appendix: Torque and fastener reference

The values below apply to fasteners that are replaced or re-torqued during maintenance of the BETA-HP-200 press. Torque is applied with a calibrated torque wrench and the value is recorded on the work order. Lubricated threads use the thread lubricant stated in the table; anti-seize paste is not used on fasteners in the hydraulic circuit.

Fastener	Size and thread	Grade	Torque	Thread condition
Frame tie-rod nut	1 5/8 in x 8	10.9	250 lbf.ft	dry, three diagonal passes
Crown and base bolt	1 in x 8	10.9	575 lbf.ft	dry
Column foot bolt	1 1/4 in x 7	8.8	450 lbf.ft	dry
Ram guide shoe bolt	5/8 in x 11	8.8	140 lbf.ft	dry
Relief valve cartridge	3/4 in x 16	8.8	45 lbf.ft	hydraulic oil film
Accumulator flange bolt	1 in x 8	10.9	235 lbf.ft	hydraulic oil film
Pump mounting bolt	1/2 in x 13	8.8	62 lbf.ft	dry

Manifold subplate bolt	3/8 in x 16	12.9	40 lbf.ft	dry
Guard panel bolt	5/16 in x 18	8.8	18 lbf.ft	dry
Light curtain bracket bolt	1/4 in x 20	8.8	8 lbf.ft	dry

Fasteners removed from a safety device are replaced, not reused. Frame fasteners are reused only if the threads are undamaged and the shank is not corroded; otherwise they are replaced with a fastener of the same grade. A fastener that has been torqued past its yield, identified by a stretched shank or a nut that turns freely after it has seated, is scrapped and the reason is recorded. The tie-rod re-torque interval is the one stated in the technical data table of the current revision of this manual.

Appendix: Glossary of hydraulic and control terms

This glossary defines the terms used in this manual. It is written for a technician who is competent in mechanical maintenance but not necessarily familiar with the press control terminology.

Accumulator: a pressure vessel that stores hydraulic energy in a compressed gas and smooths pressure peaks in the circuit.
Counterbalance valve: a valve that holds a vertical load and controls its descent, fitted in the return line from the ram.
Daylight: the open distance between the ram face and the bed with the ram fully retracted.
Die cushion: an auxiliary hydraulic circuit that supports the workpiece during forming; the BETA-HP-200 has a die cushion.
Hold-to-run: a control principle in which the motion continues only while the operator holds the command device.
Lift-off pressure: the pressure at which a relief valve first opens and passes a measurable flow.
Light curtain: an electro-sensitive protective device that stops the ram when the sensing field is broken.
Muting: the deliberate suspension of a protective device during part of the machine cycle; muting is prohibited on this press.
Pilot ratio: the ratio of the pilot pressure area to the main spool area in a piloted valve.
Pre-charge: the nitrogen pressure in an accumulator with the hydraulic side depressurised and open to tank.
Reseat pressure: the pressure at which a relief valve closes after opening; it is always below the lift-off pressure.
Safe working load: the maximum load a lifting accessory or machine may carry, as marked on the rating plate.
Zero-energy verification tag: the tag fitted after isolation and verification to record that the asset is in a de-energised and proven state.

Revision history

Revision	Effective date	Summary
Rev A	2023-01-20	First controlled issue for BETA-HP-200 after the Line 1 rebuild.
Rev B	2025-05-15	Q1 2025 incident review and 2025 reliability review. Reduced pressure limits and die entry speed, longer hold-to-run, higher minimum clamp pressure and shorter inspection intervals.